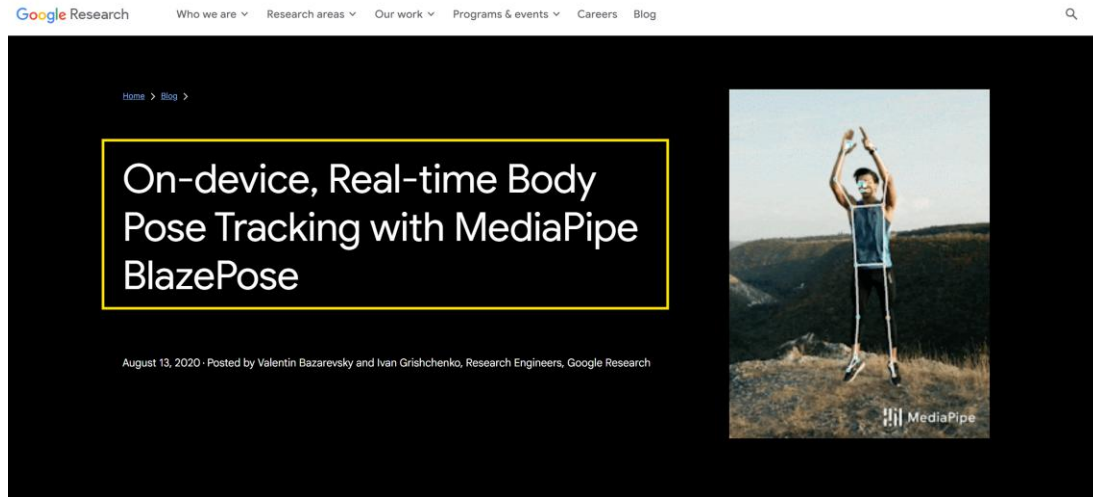
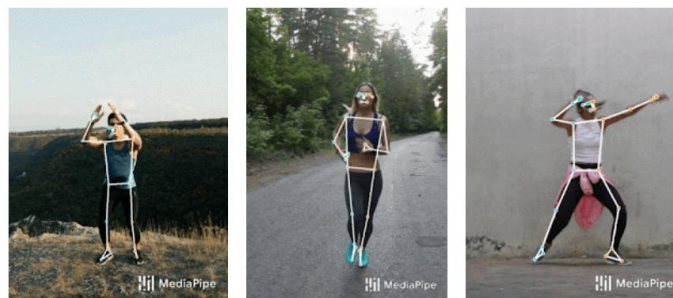


AI Model Research (MediaPipe BlazePose)

[Link to the website](#)



Pose estimation from video plays a critical role enabling the overlay of digital content and information on top of the physical world in [augmented reality](#), [sign language](#) recognition, [full-body gesture control](#), and even [quantifying physical exercises](#), where it can form the basis for yoga, dance, and fitness applications. Pose estimation for fitness applications is particularly challenging due to the wide variety of possible poses (e.g., hundreds of yoga [asanas](#)), numerous degrees of freedom, occlusions (e.g. the body or other objects occlude limbs as seen from the camera), and a variety of appearances or outfits.



BlazePose results on fitness and dance use-cases.

Today we are announcing the release of a [new approach to human body pose perception, BlazePose](#), which we [presented](#) at the [CV4ARVR workshop at CVPR 2020](#). Our approach provides human pose tracking by employing machine learning (ML) to infer 33, 2D landmarks of a body from a single frame. In contrast to current pose models based on the standard [COCO topology](#), BlazePose accurately localizes more keypoints, making it uniquely suited for fitness applications. In addition, current state-of-the-art approaches rely primarily on powerful desktop environments for inference, whereas our method achieves real-time performance on mobile phones with CPU inference. If one leverages GPU inference, BlazePose achieves super-real-time performance, enabling it to run subsequent ML models, like face or hand tracking.

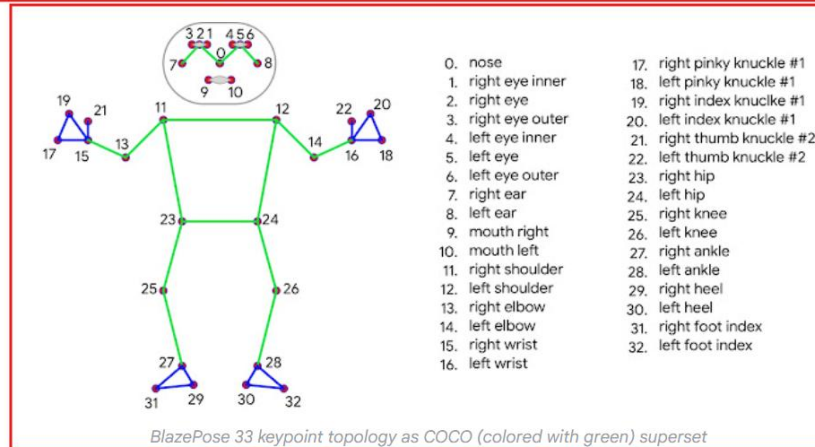


Upper-body BlazePose model in MediaPipe

Topology

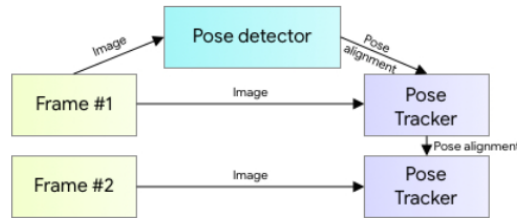
The current standard for human body pose is the [COCO topology](#), which consists of 17 landmarks across the torso, arms, legs, and face. However, the COCO keypoints only localize to the ankle and wrist points, lacking scale and orientation information for hands and feet, which is vital for practical applications like fitness and dance. The inclusion of more keypoints is crucial for the subsequent application of domain-specific pose estimation models, like those for hands, face, or feet.

With BlazePose, we present a new topology of 33 human body keypoints, which is a superset of COCO, [BlazeFace](#) and [BlazePalm](#) topologies. This allows us to determine body semantics from pose prediction alone that is consistent with face and hand models.



Overview: An ML Pipeline for Pose Tracking

For pose estimation, we utilize our [proven](#) two-step [detector-tracker ML pipeline](#). Using a detector, this pipeline first locates the pose region-of-interest (ROI) within the frame. The tracker subsequently predicts all 33 pose keypoints from this ROI. Note that for video use cases, the detector is run only on the first frame. For subsequent frames we derive the ROI from the previous frame's pose keypoints as discussed below.

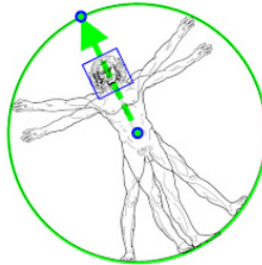


Human pose estimation pipeline overview.

Pose Detection by Extending BlazeFace

For real-time performance of the full ML pipeline consisting of pose detection and tracking models, each component must be very fast, using only a few milliseconds per frame. To accomplish this, we observe that the strongest signal to the neural network about the position of the torso is the person's face (due to its high-contrast features and comparably small variations in appearance). Therefore, we achieve a fast and lightweight pose detector by making the strong (yet for many mobile and web applications valid) assumption that the head should be visible for our single-person use case.

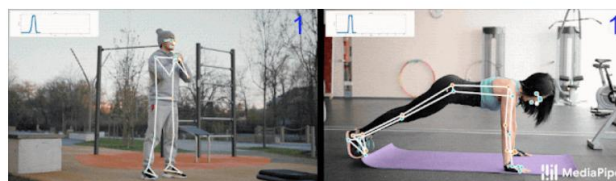
Consequently, we trained a face detector, inspired by our sub-millisecond [BlazeFace](#) model, as a proxy for a pose detector. Note, this model only detects the location of a person within the frame and can not be used to identify individuals. In contrast to the [Face Mesh](#) and [MediaPipe Hand](#) tracking pipelines, where we derive the ROI from predicted keypoints, for the human pose tracking we explicitly predict two additional *virtual* keypoints that firmly describe the human body center, rotation and scale as a circle. Inspired by [Leonardo's Vitruvian man](#), we predict the midpoint of a person's hips, the radius of a circle circumscribing the whole person, and the incline angle of the line connecting the shoulder and hip midpoints. This results in consistent tracking even for very complicated cases, like specific yoga asanas. The figure below illustrates the approach.



Vitruvian man aligned via two virtual keypoints predicted by our BlazePose detector in addition to the face bounding box

Applications

Based on human pose, we can build a variety of applications, like fitness or yoga trackers. As an example, we present squats and push up counters, which can automatically count user statistics, or verify the quality of performed exercises. Such use cases can be implemented either using an additional classifier network or even with a simple joint pairwise distance lookup algorithm, which matches the closest pose in normalized pose space.



The number of performed exercises counter based on detected body pose. Left: Squats; Right: Push-Ups